

Corruption, Social Networks, and School Cheating

Evidence from Mexican municipal audits from 2006-2013

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Abstract

Do social networks moderate the effects of corruption on exam cheating? How do rural localities differ from urban localities in their response to corruption revelations? Using a novel dataset with locality-level family network measures for rural Mexico, I provide a compelling answer to these two questions, with additional data from Mexico's municipal audits and secondary school exam cheating rates. Using a staggered difference-in-differences identification strategy with modern estimators, I find that family networks moderate the effect of corruption revelations on exam cheating rates, with a positive, non-monotonic relationship between network strength and effect size. The analysis reveals that average degree (local connectivity) drives stronger effects than largest eigenvalue (global centrality), providing suggestive evidence that peer coordination rather than information cascades explains network moderation.

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I. Introduction

When corruption scandals break in the media, do they merely inform the public, or do they fundamentally alter social behaviour? Recent evidence suggests that corruption revelations can trigger a series of behavioural changes that extend far beyond the political sphere, potentially undermining social norms and rule compliance across society.

While we increasingly understand corruption's electoral consequences, its broader social effects and spillovers remain underexplored. Research from Ajzenman (2021) shows that corruption in Mexico increased the rate of exam cheating in secondary schools, suggesting that corruption can lower rule compliance more generally. This finding is echoed by a recent experiment from Apfelstaedt & Freundt (2024), who found that when faced with corruption, people tend to be less compliant with rules in general. The implications of such a finding are wide ranging, given the importance of rule compliance for state capacity (Besley & Persson 2009, Gächter & Schulz 2016, Balán et al. 2022, Besley & Dray 2024), and the key role that a properly educated workforce has for local development, especially in regions like Latin America with severe educational inequality (Fernández et al. 2025). Although this paper does not attempt to cover these subjects, it is important to understand the broader implications of the results.

Students in rural localities are particularly affected by local government corruption, as they rely more directly on public funds for education (Ferraz et al. 2012). Furthermore, they face greater informational barriers to learn about and coordinate against corruption, given lower broadcast media competition (Larreguy et al. 2020), and lower internet access. Only 40.6% of people in rural localities use the internet versus 73.1% in urban localities (Instituto Federal de Telecomunicaciones

2019).

Nonetheless, in rural localities another informational pathway particularly stands out: social networks. Research from Arias et al. (2019) showcased how social networks moderate the effect of information about corrupt behaviour in local government in Mexico, with voters in stronger networks coordinating against malfeasant incumbents. Yet, no research has so far explored the way social networks moderate the effect of corruption revelations on non-electoral forms of behaviour, such as cheating in exams, despite its clear implications for development.

In this dissertation, I study whether social networks moderate the effect of corruption on the cheating behaviour of secondary students in rural Mexico. To tackle this question, I make use of a novel dataset with locality-level family network measures for over 10,000 rural Mexican localities and nearly 150,000 grade-school-year observations. To obtain causal estimates, I employ a staggered difference-in-differences identification strategy with modern estimators. In my main results, I identify a positive, strong, and statistically significant effect of corruption on cheating rates in rural Mexican localities (1.9-2.9pp.), but only in those localities with strong social networks, highlighting the importance of family networks in rural Mexico as both information transmission and coordination mechanisms.

This dissertation makes three key theoretical contributions:

1. First, I demonstrate that social networks are not merely conduits but active moderators of corruption's behavioural effects. While Ajzenman (2021) documents average effects, I show these effects are contingent on social structure, with corruption's impact on cheating ranging from null to a striking 55% increase depending on network strength. This reveals that previous estimates have masked crucial heterogeneity.

2. Second, I provide evidence that this moderation operates primarily through local peer-to-peer coordination rather than information cascades. A distinction revealed by comparing average degree and largest eigenvalue effects, this finding could have profound implications for understanding how corruption norms spread.
3. Third, I show that in information-constrained rural environments, strong family networks can actually amplify corruption's behavioural effects, not only mitigating but reversing the dampening effect of weak information transmission found in rural contexts. These findings advance our understanding of how informal institutions shape responses to formal institutional failures.

II. Literature Review and Theory

This dissertation examines how social networks moderate the effect of corruption revelations on academic cheating behaviour in rural Mexican communities. Given the novel nature of this research question, I draw from literature on corruption and electoral accountability, social networks and information transmission, and compliance behaviour to build a coherent theoretical framework. The review is structured to first establish the baseline effect of corruption revelations on behaviour, then examine how social networks moderate information transmission and norm compliance, before synthesizing these insights into an original theory.

A. *Corruption and Behavioural Responses*

The literature on electoral accountability is foundational for understanding how corruption revelations affect citizen behaviour. Retrospective voting models predict that voters will sanction corrupt incumbents once they receive information about malfeasance or low competence (Ferejohn 1986, Persson et al. 1997, Ashworth 2012). However, empirical evidence is mixed regarding whether this electoral sanctioning actually occurs or not (De Vries & Solaz 2017, Dunning et al. 2019). Empirical evidence from Brazil's randomized municipal audits confirms this baseline prediction (Ferraz & Finan 2008, 2011, Avis et al. 2018), as they find that corruption revelations reduce incumbent vote shares, but emphasise that sanctioning only occurs when information is available and formal accountability institutions are strong.

Some level of accountability following corruption has also been documented by Berliner & Wehner (2022) with South African audits, Larcinese & Sircar (2017) in the UK, Costas-Pérez et al. (2012) in Spain, and Larreguy et al. (2020) and Enríquez et al. (2024) with Mexican audits. These stud-

ies reinforce the broader finding that information remains a key lever for electoral sanctioning (Ashworth 2012).

On the other hand, in Mexico as well, Chong et al. (2015) found that corruption revelations decrease support for both incumbents and challengers, while reducing overall turnout. Similarly, Arias et al. (2022) show that electoral sanctioning in Mexico is limited when voters hold unfavourable prior beliefs about government integrity, with corrupt incumbents sometimes even gaining vote share. Boas et al. (2019) suggest that while voters oppose corruption in principle, their behaviour is constrained by a misallocation of responsibility regarding corruption that is superseded by more pressing concerns at the municipality-level such as healthcare and employment.

These findings highlight that prior beliefs about corruption moderate behavioural responses, and that rural communities may experience less effective information flows due to less competitive media and weaker formal institutions.

In Mexico's rural localities, there exists an additional potential issue regarding corruption and sanctioning, which is that some municipalities revealed as corrupt may have not only known about the corruption scandal, but may have participated and benefitted from it. This touches on a strand of literature which has identified that electoral sanctioning is limited when corruption is seen to have welfare-enhancing rather than welfare-decreasing effects (Fernández-Vázquez et al. 2016). Although the welfare effects of corruption are difficult to estimate at an individual level, it is likely that those participating in the scandal perceive them as welfare-enhancing, and thus fail to sanction (or react to) it.

More directly relevant to this study, Ajzenman (2021) provides evidence that corruption affects rule compliance beyond electoral behaviour. Using the same data as this dissertation, Ajzenman

finds a positive association between municipal corruption and student cheating rates, with stronger effects in areas with more competitive media. This establishes the baseline expectation that corruption revelations increase rule violations in educational settings. Nonetheless, in his study he provides little theorisation of whether social networks continue moderating the effect, and whether the magnitude and significance of the effect are sustained in rural localities, which have less effective information flows. Hence, it is now important to assess whether networks can mitigate the lost information transmission effectiveness.

B. Social Networks and Information Transmission

Given the informational challenges in rural communities, it becomes crucial to understand how alternative information channels operate. Social networks play a particularly important role in both information transmission and behavioural coordination in contexts where media competition and institutions are weak. Nonetheless, the biggest challenge for researchers studying the effects of social networks is mapping them accurately (Chandrasekhar & Lewis 2016). Empirical social network work presents many researcher-degrees-of-freedom, given the multiple characterisations of social ties, distance, and aggregate measures available to researchers.

One popular social network measure (proxy) is Facebook’s Social Connectedness Index (SCI), used by Enríquez et al. (2024) in Mexico, Mooers (2024) in the US, and first introduced by Bailey et al. (2018). In essence, the SCI measures how often Facebook users interact with users in another location, and thus how often users interact with other users in their own location. The similarity of these online social ties to offline relationships in political contexts like the one studied here seems to be strong (Bisbee & Larson 2017)

Yet, in contexts like rural Mexico, social network measures built with administrative datasets seem a better proxy. This is primarily due to the lower rate of individuals with access to social media, nearing 30% in rural localities versus 57% in urban localities, as of 2019 (Instituto Federal de Telecomunicaciones 2019).

This motivates the use of another popular proxy of social network strength: family ties. In rural Mexico, the family is a key source of political information, with 47% of survey respondents in the Arias et al. (2019) experiment indicating to have discussed politics with their household members in the week prior to the interview. Furthermore, family networks in Mexico have been shown to have an effect on other forms of individual behaviour, such as the incentives to migrate (McKenzie & Rapoport 2010), and the flow of remittances (Woodruff & Zenteno 2007).

In particular, two measures of social network centrality stand out: average degree and largest eigenvalue. In essence, the average degree network measure represents the average number of people that an individual in a locality is connected and can coordinate with within that same locality. The largest eigenvalue measure represents the extent to which an individual is central within a locality, thus how well-connected that individual is to other well-connected individuals in that locality. For a more intricate explanation of these two measures see Alatas et al. (2016) and Arias et al. (2019).

Average degree captures direct connectivity while largest eigenvalue measures recursive centrality. Significant average degree effects, without significant eigenvalue effects, would suggest cheating increases primarily through immediate peer influence rather than through deeper network cascades, indicating that the corruption norm transmission operates at the local neighbourhood level rather than through the broader community's most central actors. Alternatively, if largest eigenvalue effects are significant but average degree effects are not, this would indicate that corruption norms

spread primarily through influential, well-connected individuals who serve as key conduits for behavioural change, suggesting that norm transmission requires central actors rather than simple peer-to-peer influence.

Previous research on the role of social networks in the educational outcomes of children in Mexico already highlighted how centrality measures were the best predictors of academic performance and behaviour (Ramírez Ortiz et al. 2004). Furthermore, in Arias et al. (2019), it is only the average degree and largest eigenvalue measures which closely reflect the real social networks of field experiment participants in rural Mexico, and the only network measures which exhibit a high correlation between them (93% in my sample, and 98% in the Arias et al. (2019) sample).

For information transmission specifically, Simpser (2020) documents intergenerational persistence of corruption attitudes within Mexican families, indicating that family networks are primary channels for transmitting norms about corruption. This finding is particularly relevant for understanding how corruption revelations reach secondary school students, who are unlikely to consume local news independently but receive information through family discussions.

Fowler & Christakis (2010), Chandrasekhar et al. (2018) and Chandrasekhar et al. (2020) provide two key insights for understanding network effects. First, social networks can substitute for formal institutions in enforcing cooperation, though their effectiveness decreases with social distance. Second, networks contain different types of learners, and in communities with lower educational attainment, information transmission may fail even when coordination mechanisms remain intact, given the higher chances of encountering DeGroot learners (who update beliefs via weighted averaging across the social network) rather than Bayesian learners (who update beliefs by conditioning on new information using Bayes rule). The implications of this characteristic of networks for our

study are multiplicative given the importance of education for the discernment of the quality of information on corruption (Weitz-Shapiro & Winters 2017), and the weaker learning of political information from social transmission versus media sources (Carlson 2019). Thus, we can expect that on average rural localities will be less reactive to corruption revelations than the generally more educated and media-informed urban communities.

Cruz (2019) and Cruz et al. (2020) extend this analysis of (family) networks to political contexts, supporting the idea that information transmission effectiveness often depends on network structure, while confirming that these general functions of networks are applicable in developing contexts like rural Mexico, even if they present countervailing effects such as lower educational attainment (Weitz-Shapiro & Winters 2017). However, how can networks moderate the behavioural spillovers of corruption such as exam cheating?

C. Compliance Behaviour and Social Coordination

The literature on compliance behaviour reveals that rule violations are fundamentally social phenomena (Keizer et al. 2008, Apffelstaedt & Freundt 2024). Gino et al. (2009) finds that exam cheating is driven not only by cost-benefit calculations but by social norms implied by others' dishonesty and the salience of dishonest behaviour. This is something that has also been found in studies of corruption, like that from Fisman & Miguel (2007). When corruption revelations increase the salience of dishonesty, we expect increased student cheating, similar to how we expect a level of sanctioning in electoral accountability settings, echoing a more general model of reciprocity like that proposed by Falk & Fischbacher (2006).

Furthermore, as indicated by Wolitzky (2013), Larson (2017) and Cruz et al. (2020), social net-

works often lead to higher coordination when the penalty for free-riding or non-cooperative behaviour is greater. In the example of school cheating, it is clear why secondary school students would want to sanction non-cooperative classmates, who could call them out, as a form of retaliation (Falk et al. 2005).

The social nature of compliance is further supported by research on tax evasion. Torgler (2002), Alstadsæter et al. (2019), and Gamannossi degl’Innocenti & Rablen (2020) all find that individuals evade more taxes when they know others who evade, suggesting that the effects of the network operate through both social norms and information transmission channels (Cruz et al. 2020).

While Ajzenman (2021) showed how the effects of corruption revelations on cheating rates varied by media connectivity and level of corruption revealed, educational behaviour is much more closely determined by family and peer influence. This is not to say that media and level of corruption do not matter, but that family/network influence is an essential step in driving behavioural effects in children.

Research on student cheating specifically supports this social coordination mechanism. Carrell et al. (2008), Dee & Jacob (2012), Fischbacher & Föllmi-Heusi (2013) and Zhao et al. (2022) find that cheating behaviour spreads through peer and family networks, with students more likely to cheat when they observe or expect others to cheat, or when their parents seem to support such practices of academic dishonesty. In particular, Ramírez Ortiz et al. (2004) already explored the relevance and impact of social networks in the educational outcomes of Mexican children, and in particular in the context of poverty, a common occurrence in rural localities. This suggests that the mechanisms driving voter coordination following corruption revelations (Arias et al. 2019), or voter coordination in general (Finan et al. 2021), may operate similarly in educational settings, and

that family networks will be amongst the most important factors driving exam cheating.

D. Theory

These three bodies of literature converge to suggest that understanding corruption's impact on cheating requires examining how social networks moderate both information flows and behavioural coordination. Combining these insights, I propose a theory where corruption revelations affect student cheating through two mechanisms moderated by social network strength. This reflects the "two-step flow of communication" (Enríquez et al. 2024). My theory operates through three sequential stages:

Stage 1: Information Transmission Municipal audits reveal corruption in a subset of municipalities each year. When corruption is revealed, this information reaches secondary school students primarily through family and social networks rather than through direct media consumption, as documented by Simpser (2020) and consistent with the "two-step flow of communication" literature. The effectiveness of this transmission mechanism depends critically on network strength. In localities with strong family networks, corruption revelations spread more effectively due to multiple reinforcing communication channels within densely connected family structures. These networks facilitate both the initial transmission of audit findings and their subsequent amplification through repeated family discussions, as observed in Arias et al. (2019) field experiments where family members proved to be primary sources of political information. Conversely, in localities with weak family networks, information transmission faces substantial barriers: fewer communication pathways mean corruption revelations may never reach students, or arrive in fragmented, less credible forms. This differential information penetration based on network strength is par-

ticularly pronounced in rural Mexico, where formal media competition is limited (Larreguy et al. 2020) and family networks serve as crucial alternative information channels. The likelihood of students receiving and processing corruption information thus varies systematically with their locality's network characteristics, creating the foundation for heterogeneous behavioural responses.

Stage 2: Norm Updating Upon receiving corruption information, students update their beliefs about the acceptability of rule violations. Corruption revelations are expected to increase the salience of dishonesty (Gino et al. 2009, Zhao et al. 2022) and signal that rule violations are widespread, reducing the perceived costs of cheating. This updates student priors before they decide whether to cheat. As highlighted in various papers, corruption and rule non-compliance are heavily influenced by social and family norms (Fisman & Miguel 2007, Simpson 2020, Besley & Dray 2024).

Stage 3: Behavioural Response Students coordinate their cheating behaviour based on updated norms and expectations about others' behaviour. In stronger family networks, this coordination is more effective due to better information about peers' intended behaviour and stronger enforcement of group decisions (coordination) through penalties and potential retaliation (Falk et al. 2005, Chandrasekhar et al. 2018).

Network Moderation Effect Strong networks amplify the corruption-cheating relationship through two mechanisms: (1) more effective information transmission ensures more students receive corruption signals, and (2) better coordination capabilities allow students to act collectively on updated norms. This leads to the **primary hypothesis** that the positive effect of corruption revelations on cheating behaviour will be stronger in municipalities with stronger social networks.

This theory accounts for the rural Mexican context where formal institutions are weaker (Chong

et al. 2015), family networks are primary information channels (Simpser 2020), and prior beliefs about corruption are high (Arias et al. 2022). The network moderation effect is particularly relevant in this context because family networks may be the primary mechanism through which corruption information reaches secondary school students and influences their behavioural responses.

III. Data

A. *Cheating-Audit Data*

Both the Mexican municipal audit and school cheating datasets were obtained from the Ajzenman (2021) replication package. These two datasets range from 2006-2013, as these are the only years with valid observations of cheating and corruption. After 2013, the national, standardised exam "*Prueba ENLACE*" was replaced by "*Prueba PLANEA*", limiting the dataset to these years. The main outcome tested in this paper is the proportion of students that cheated on the "*Prueba ENLACE*", a national, standardised exam taken each year between mid-April and mid-July. This test was administered to primary school kids between third and sixth grade (~8-11 years old), and secondary school students in their third year of secondary school (~14-15 years old), with the exception of some first- and second-year students who also took it in 2009.

To calculate the proportion of cheaters, the Secretaría de Educación Pública de Mexico (Mexican Ministry of Public Education, or SEP) used two statistical algorithms (i.e., the K-Index and Scrutiny Method), also used by the organisers of standardised tests like the GRE, TOEFL and SAT, to identify rates of student-to-student cheating. These algorithms identify patterns of incorrect answers between any pair of tests within a classroom, and are the most common and court-defensible methods for identifying cheating rates (Cizek 1999). Nonetheless, it is important to note that this measure does not capture teacher-student cheating, for which there was no readily available data. The exam cheating dataset also includes variables for the grade of the students, their shift (morning, afternoon, night), a public/private school indicator, and classroom size.

On the other end of the equation, the corruption data comes from the municipal-level, public fund

audits carried out by the Auditoría Superior de la Federación (Federal Audit Office of Mexico, or ASF). These audits have been carried out on an annual basis since the year 2000, though with substantial changes after 2016, when the National Anti-Corruption System (Sistema Nacional Anticorrupción, or SNA) was implemented. The corruption variable used particularly focuses on budgetary deviations from the FISM (Municipal Fund for Social Infrastructure) which, as the name indicates, are funds earmarked to improve social infrastructure projects which primarily benefit the poor, thus any "unauthorised use" of these funds is expected to receive substantive media coverage. The FISM represents over 25% of a municipality's budget. This approach is "inherited" from Chong et al. (2015), Larreguy et al. (2020), Ajzenman (2021) and Arias et al. (2022), who also use deviations from the FISM as an objective measure of corruption, following the example from Brazil's own audits (Ferraz & Finan 2008, 2011).

If the audits report any unauthorised use of FISM funds, my corruption (treatment) variable equals 1, otherwise it equals 0. Over the sample, 930 audits were carried out (117 per year), with 54% of them reporting some degree of corruption, and in those corrupt municipalities, an average of 8% of funds used were classified as "unauthorised", as reported by Ajzenman (2021). Over the sample, a quarter of all Mexican municipalities were audited.

The audit results used in this dissertation were released each February, one year after the start of the investigation and two years after any unauthorised use could have occurred, which is around 2-3 months before students are scheduled to take their school exams. These audit reports were quite popular, and this is evidenced through Figure 1 from Ajzenman (2021) which shows how Google searches of the word "audit" ("auditoría" in Spanish) spike in February for each of the years in my sample.

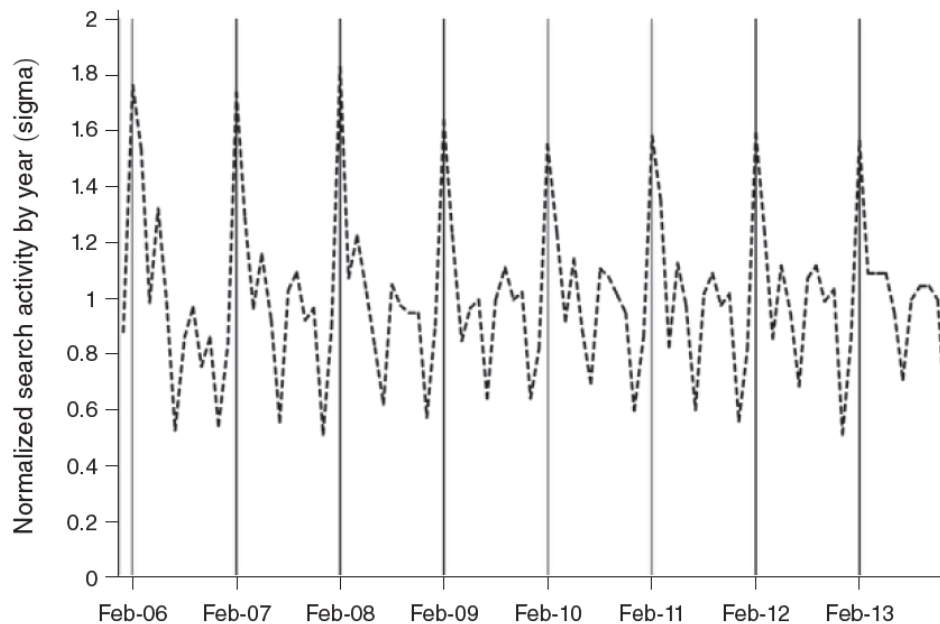


Figure 1: Google Search history of word "auditoría" in Mexico

Notes: Figure obtained from Ajzenman (2021), who previously extracted it from Google Trends, normalising the values by their yearly mean.

B. Network Data

The locality-level¹ network data was constructed using the full list of Prospera beneficiaries, obtained from the Arias et al. (2019) replication package.

The network data was constructed with administrative data from an almost complete list of Prospera beneficiaries in 2017². Prospera is a conditional-cash-transfer social programme in Mexico.

The list of individual Prospera beneficiaries and their localities can be obtained from datos.gob.mx.

While the Prospera programme is now defunct after being eliminated in 2019, at its peak before elimination it served approximately 6.8 million families, which represents around 25-30% of all

¹Localities are administrative subdivisions within municipalities. In Mexico, as of 2025, there are 296,854 localities, with 291,000 classified as rural.

²Some irregularities in the tracking of Prospera beneficiaries have been reported, thus it is possible some beneficiaries were not present in the list (Dávila Lárraga 2016). In particular, it is noted that there is no data on Prospera beneficiaries for Durango, likely due to lateness from the entity's reporting body. Future studies should attempt to include Durango to see if the effect holds.

Mexican households. It was particularly present in rural areas, where it covered roughly 40-50% of all rural households. Some of the conditions to receive the cash transfers included regular school attendance, regular health check-ups at designated clinics, and attending nutrition workshops. The cash transfers were typically given to the women in the household.

These characteristics about the Prospera programme have many important implications for this study, given the network data used is built with it. First off, some rural localities did not have data on any Prospera beneficiaries, thus no locality-level network measure exists for them. Of the 296,854 localities in Mexico, and 116,000 which received Prospera cash transfers (Dávila Lárraga 2016), only 66,061 localities had sufficient Prospera beneficiaries in the dataset to construct a reliable network measure (see sample restrictions below). From the 66,061 localities for which I constructed network data, only 10,080 had schools present within them. The Prospera beneficiary dataset, before any sample restrictions, had data for 112,357 unique localities, and 12,088,415 entries of individual Prospera beneficiaries.³

It is likely that the rural localities with Prospera beneficiaries in the dataset were systematically different from the rural localities without Prospera beneficiaries in the dataset. It may be that in especially low-capacity localities, no Prospera beneficiaries were listed because the conditions imposed for the cash transfer were implausible to keep for certain families. Thus, my sample likely reflects a low-state-capacity sample of rural localities in Mexico which received substantial help from the Prospera programme, but still present slightly better infrastructure than the poorest rural localities of Mexico which have no schools. The additional mismatch between localities

³This does not match the entire population of Prospera beneficiaries or number of households receiving benefits, as some families just had one individual receiving the cash transfer, while others had multiple members claiming the transfers.

with Prospera beneficiaries and my sample with network data is due to the lower (>10) and upper ($<2,500$) limits of population imposed when constructing the dataset, plus the non-existence of schools in most rural localities. Future studies would likely benefit from incorporating geospatial data to account for those school children who attend school outside their locality boundaries, a common circumstance that impacts at least 55,000 rural localities.

The construction of the locality-level network data was done in a similar way to the Arias et al. (2019) precinct-level network data. The main difference from their approach and mine is that my main results use families as nodes given their greater computational efficiency, although alternative specifications can use individuals as nodes, like Arias et al. (2019) do for their main results.⁴ My approach also differs from theirs in that my social network measures are aggregated at the locality- rather than precinct-level. This was done to avoid further mismatches between precincts with and without schools, as localities are often larger than voting precincts, and because there is no theoretically-informed reason to aggregate the network data by precinct for this dissertation, as our outcome is not related to electoral performance. Arias et al. (2019) note that their main results held when using families as nodes, and given the aggregation of the network data at the locality level, any non-classical measurement error (Wang et al. 2012, Chandrasekhar & Lewis 2016) is likely reduced.

In Appendix B, I have included graphs with the distribution of the average degree and largest eigenvalue measures, a correlation graph of average degree and largest eigenvalue, and a distribution of networks sizes. The correlation between my average degree and largest eigenvalue measures is

⁴The processing time for using individuals as nodes to construct the locality-level network measure would have been >24 hours, compared to the 3 hours that the approach using families as nodes takes on average. Replication materials available in Appendix A.

93%, sufficiently close to the 98% correlation observed by Arias et al. (2019). The difference in correlation between measures may be due to the larger list of surnames that I worked with to create the network measures relative to the smaller number of individuals in the Arias et al. (2019) experimental sample, despite surname cleaning.⁵

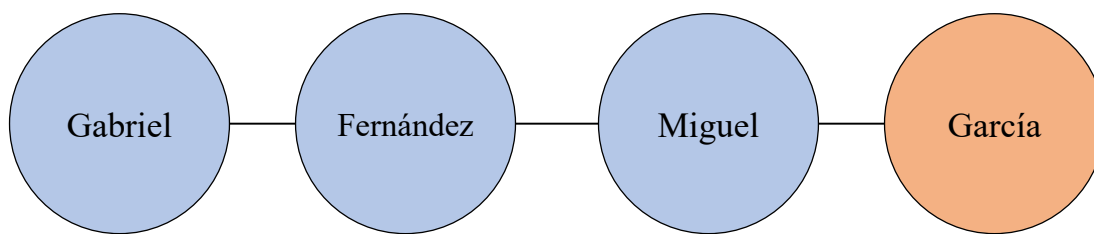
Conceptually, the average degree network measure is the mean number of connections per node (family) across the entire network (locality). The largest eigenvalue (of the adjacency matrix) measure, on the other hand, captures the extent to which the average family unit is central in the sense that it is connected to other family units that are recursively deemed to be highly central within the locality's social structure. Their key difference lies in how average degree captures more closely local connectivity with a focus on first-order connections, while largest eigenvalue captures a more "global" network structure with a focus on higher-order connectivity effects (i.e., the connectivity of your connections).

For clarification, in Mexico and other Spanish-speaking countries, people typically retain both of their parents' last names. The use of administrative data for network construction is preferred over sampled network data (Chandrasekhar & Lewis 2016), and is becoming increasingly common given its lower measurement error and bias relative to sampled networks (Arias et al. 2019, Alt et al. 2022).

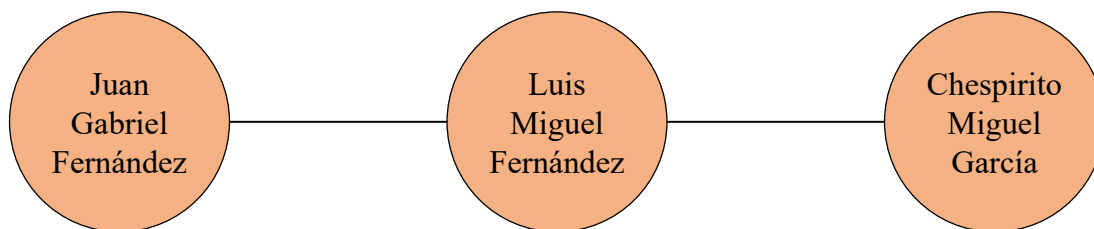
When denoting a family as a node, each unique family name (surname) becomes a single node in the network. As an example, see Figure 2a. If a Prospera beneficiary is called Luis Miguel Fernández, he directly connects family names Miguel and Fernández, while Juan Gabriel Fernán-

⁵Arias et al. (2019) worked with network data for 95,199 Prospera beneficiaries. For contrast, this dissertation worked with network data for Prospera beneficiaries in 66,061 localities, or over 7,000,000 individual beneficiaries. The total population of the localities for which I created network data is 23,448,149, or 19% of the entire Mexican population, as of 2017.

dez directly connects the family names Gabriel and Fernández. Nodes Miguel and Gabriel are indirectly connected to each other through the surname Fernández. If I were using individuals as nodes instead, each individual Prospera beneficiary becomes a separate node in the network. In this example, see Figure 2b, Luis would connect to any other individual who shares Miguel or Fernández as one of their surnames, while Juan would connect directly to any individual sharing Gabriel or Fernández. If there was a third person called Chespirito Miguel García, then Luis would be directly connected to Chespirito, and Juan would be indirectly connected to Chespirito through Luis.



(a) Families as nodes



(b) Individuals as nodes

Figure 2: Example of Network Links between Prospera Beneficiaries

This approach is a methodological improvement over sampled network data as indicated by Chandrasekhar & Lewis (2016) and Cruz et al. (2020). This is because sampled network data often

introduces systematic biases that are difficult to correct. As indicated above, aggregation at the locality-level should further decrease the chances of non-classical measurement error in network construction.

C. Sample Restrictions and Merging

It is important to clarify that the network data and construction method used here only makes sense at the locality-level, rather than at the municipality-level. The main reason for this is that the locality-level more closely reflects real ties than municipalities given the greater geospatial precision of administrative subdivisions. While in a municipality it is unlikely that all of those with the surnames Gabriel and Fernández are related to each other, at the locality-level the probability that they are indeed related is much greater. Still, not all localities are the same in terms of social-tie-matching precision, in fact, my network data excludes urban localities. Measured ties in urban settings would likely suffer from the same social-tie-matching imprecision bias of municipalities.

The main reason for this is that rural communities, as Arias et al. (2019) indicated, are more tight-knit and experience lower levels of migration than urban localities, meaning common surnames in a rural area are more likely to be an indication of a genuine (rather than spurious) family tie. What is meant by a genuine versus spurious social tie is that we want network data to reflect real family ties. Thus, I want my measures of social ties to capture the real connections of people and families, so my theory of information transmission and coordination can be tested for. Conversely, urban localities include a larger proportion of individuals who share surnames out of random chance due to the commonness of certain last names like 'García' or 'Hernández', but have no real connection between them. This focus on rural localities represents a deliberate methodological choice

that strengthens causal identification. This rural focus thus prioritises measurement validity over sample size, a critical trade-off for causal inference.

Given my network data is built from a list of Prospera beneficiaries, with conditionality for participation, measured ties more closely reflect real connections between individuals sharing last names in rural localities than in urban settings. This is due to what is discussed in the literature review section on information transmission in rural settings, where participation into social programmes like Prospera due to word-of-mouth information transmission between families is more common in rural settings where alternative information sources are not as present.

The upper population bound I used to construct the network data is 2,500, the threshold for urban-rural classification. To avoid injecting severe measurement error into my average degree and largest eigenvalue measure, as reported by Wang et al. (2012) and Chandrasekhar & Lewis (2016), I only created network measures for those localities which had at least 10 beneficiaries in the Prospera dataset. While this threshold of 10 is arbitrary, future research could explore lower and higher thresholds for robustness. The lower bound of >10 Prospera beneficiaries eliminated over 40,000 localities from the Prospera beneficiary dataset, given the total number of rural localities in the dataset was 108,394 before subsetting for those with a population above 10. In Table 1, we can see some more data on the number of schools, localities, municipalities and states (Entidades) for which there is network data, and how each of these administrative subdivisions are represented by the data.

The merging of data was done using unique locality identifiers. As mentioned above, despite having network data for 66,061 localities, only 10,080 localities had schools within them to match with the school cheating and audit data. This is not entirely surprising as rural Mexican localities

Table 1: Network Coverage Statistics

State	Unique Localities	Unique Schools	Unique Municipalities
<i>Total</i>	10,080	10,384	533
Aguascalientes	160	157	10
Baja California	81	82	5
Baja California Sur	59	60	5
Campeche	214	223	11
Coahuila	86	94	13
Colima	60	64	9
Chiapas	892	900	37
Chihuahua	217	234	13
Guanajuato	761	782	26
Guerrero	864	913	33
Hidalgo	435	437	26
Jalisco	158	175	14
Mexico	523	531	32
Michoacan	403	422	28
Morelos	59	65	16
Nayarit	301	308	11
Nuevo Leon	125	124	9
Oaxaca	290	312	24
Puebla	394	406	30
Queretaro	231	235	14
Quintana Roo	150	154	8
San Luis Potosi	626	655	19
Sinaloa	529	525	15
Sonora	232	244	15
Tabasco	466	464	15
Tamaulipas	188	189	13
Tlaxcala	68	74	12
Veracruz	716	733	32
Yucatan	167	172	20
Zacatecas	625	650	18

Notes: Statistics calculated for all observations where there is network data. The total population covered is 6,852,803, or 5.5% of the entire Mexican population and 26.3% of the entire rural population. There are no observations for Durango (due to lack of data in Prospera beneficiaries dataset) or Mexico City (with no rural localities).

have low state capacity and the presence of a school is quite rare (Ramírez Ortiz et al. 2004). Students often have to go to neighbouring localities, or even municipalities, to attend school, which is one of the reasons for the unequal educational system in Mexico (Fernández et al. 2025) and high dropout rates in rural communities (Ramírez Ortiz et al. 2004).

The final sample size is 148,842 grade-school-year observations.⁶ As mentioned throughout the data section, several limitations may create certain biases in the results, particularly given that geospatial data on neighbouring localities could not be incorporated to better accommodate real networks around school locations, rather than arbitrary locality-level measures which may not contain a true sample of the children attending school "X". It is likely that at school "X" there are students from various localities with various, even if similar, network strengths.

While the localities for which there is data are likely representative of rural localities in Mexico, there may also be some bias generated from the network measures used. While the approach outlined above is expected to be a robust methodology that will somewhat closely reflect real social ties, it may also be that non-classical measurement error remains (Wang et al. 2012, Chandrasekhar & Lewis 2016). The aggregation of network data at the locality-level should mitigate some of these concerns, and there is some evidence that suggests that any spurious ties captured by an aggregated network measure should not be correlated with political or compliance behaviour (Cantú 2014). Despite these limitations, this approach represents a methodologically sound attempt to measure social networks in rural Mexico using available administrative data.

Table 2 confirms that our dataset, relative to Ajzenman's remaining sample (i.e., observations without network data), has generally poorer municipalities, with lower radio presence rates, higher

⁶In 2009, multiple grades took the "*Prueba ENLACE*", thus they are separated

levels of marginality, higher base exam cheating rates, a lower proportion of private schools, and, interestingly, a lower probability of prior revealed corruption, and lower unauthorised spending in the audits.

Table 2: Summary Statistics by Network Presence

	No Network (N = 192,790)		Has Network (N = 148,842)		
Variable	Mean	(SD)	Mean	(SD)	Difference
<i>Municipality Indicators</i>					
Logged GDP per capita	11.522	(0.516)	11.443	(0.537)	-0.079
Radio presence rate	0.650	(0.113)	0.527	(0.150)	-0.123
Level of marginality	1.882	(1.067)	3.796	(0.744)	1.914
<i>Education Indicators</i>					
Exam cheating rate	0.031	(0.087)	0.053	(0.160)	0.022
Private school (=1)	0.247	(0.431)	0.004	(0.061)	-0.243
<i>Political Indicators</i>					
Same party as federal govt (=1)	0.336	(0.472)	0.290	(0.454)	-0.046
Same party as state govt (=1)	0.561	(0.496)	0.499	(0.500)	-0.063
Prior corruption (=1)	0.541	(0.498)	0.381	(0.486)	-0.160
Unauthorized spending (%)	2.465	(8.997)	1.912	(7.220)	-0.553

Notes: Standard deviations in parentheses. Difference column shows the mean difference between groups (Has Network - No Network). For binary variables, means represent proportions. The level of marginality variable is an ordered categorical variable converted to numerical values for five levels (very low = 1, low, medium, high, very high = 5) of marginality.

IV. Methodology

The identification strategy used in this paper is a staggered Difference-in-Differences (Diff-in-Diff) design. This identification strategy differs from traditional, two-period Difference-in-Differences given treatment is assigned in different time periods across units, though the key identification assumptions remain roughly the same.

To interpret these coefficients as causal we require meeting the key identification assumptions from staggered Diff-in-Diff: parallel trends, and no anticipation effects (Baker et al. 2025). Parallel trends assumes that treated units (i.e., those municipalities which are audited AND are revealed as corrupt) would have followed a similar trend in exam cheating rates as control units (i.e., those municipalities which are audited AND are NOT revealed as corrupt). The no anticipation identification assumption from Diff-in-Diff is in essence already covered by parallel trends, but particularly refers to the lack of anticipatory effects from treated units in periods preceding the date of treatment assignment with regards to exam cheating rates.

The theoretical justification for parallel trends in this study is that localities which are audited and revealed as corrupt would likely follow similar trends in exam cheating rates to localities which are also audited but are not revealed as corrupt, had they not mismanaged FISM funds. By restricting my analysis to rural-only localities, I improve the theoretical justification for parallel trends, thanks to the increased comparability between treated and control units. While limiting my external validity, by restricting my dataset I improve this study's internal validity for causal inference, as treated and control units are more likely to experience shocks in a similar manner, such as agricultural or fiscal reforms.

As is common with most causal identification strategies with observational data, neither of these two identification assumptions can be directly tested for, but several diagnostic falsification tests exist which can give some indication of whether they are likely met or not.

The most popular falsification tests for parallel trends are event-study plots, which visually represent the differences in the outcome (i.e., exam cheating rates) between treated and control units in the years preceding and following treatment assignment (i.e., the year the audit results are posted online). If met, these parallel *pre-trends* tests, while not directly proving or disproving the parallel trends assumption, are informative and give us some confidence in the validity of the results.

Recent econometric literature highlights that the traditional two-way fixed effects (TWFE) estimator can be biased in staggered settings due to problematic comparisons between treated and control units (Callaway & Sant'Anna 2021, Sun & Abraham 2021, Goodman-Bacon 2021, Imai et al. 2023, de Chaisemartin et al. 2024, Liu et al. 2024, Baker et al. 2025, Gardner et al. 2025). To address this, I use the robust estimator from Sun & Abraham (2021) for my main results, but will also employ the diagnostic tests provided by Liu et al. (2024) to assess the parallel trends assumption.

Flexible matching and re-weighting estimators, such as those from Sun & Abraham (2021) and Callaway & Sant'Anna (2021), solve the "forbidden comparisons" problem by making the "right" comparisons only, weighting the $[cohort * time * treatment]$ estimates appropriately, and then recovering the ATT. Counterfactual imputation estimators such as Liu et al. (2024), Borusyak et al. (2024) and Gardner et al. (2025), on the other hand, can be generally described as modelling untreated potential outcomes $Y(0)$ using only untreated observations, and then using these to estimate (impute) counterfactual untreated outcomes for treated observations. One of the slight disadvan-

tages of counterfactual estimators, such as the interactive fixed effects counterfactual (IFEct) and matrix completion (MC) methods, is that these require a large amount (>5) of untreated periods to estimate the ATT, thus they may be biased in this dissertation's sample given the timeline I am working with (2006-2013) would make only units treated in 2011 and 2012 methodologically apt for these methods, significantly reducing the sample size.

My empirical specification is:

$$Y_{g,s,t}^{(q)} = \kappa_{grade}^{(q)} + \alpha_{school}^{(q)} + \delta_{muni}^{(q)} + \lambda_t^{(q)} + \beta D_{muni,t}^{(q)} + \tau \mathbf{X}_{muni,t}^{(q)} + \epsilon_{school,t}^{(q)}, \quad q = 1, 2, 3, 4, 5 \quad (1)$$

Notes: Equation 1 is estimated separately for each quintile ($q \in \{1, 2, 3, 4, 5\}$) of the distribution of average degree (network) centrality, using families as nodes. All coefficients are allowed to vary by quintile.

In my specification, $\kappa_{grade}^{(q)} + \alpha_{school}^{(q)} + \delta_{muni}^{(q)} + \lambda_t^{(q)}$ refer to the grade, school, municipality and time fixed effects. This allows us to focus this study on the within-unit changes in the proportion of exam cheating following the revelation of municipality-level corruption, captured by the treatment variable $D_{muni,t}^{(q)}$. The covariates $\mathbf{X}_{muni,t}^{(q)}$ included in the main specification are the same as those included in Ajzenman (2021), given their equal applicability to this study. These are whether a report was released that year, what turn (morning, afternoon, night) the students attended, the homicide rate by municipality, whether the municipality had been revealed as corrupt in the past (as an attempt to control for priors), and whether the local government was politically aligned with the federal government.

By controlling for these variables, we ensure that parallel trends remains more plausible, but this slightly changes and relaxes our assumption from unconditional parallel trends to *conditional* parallel trends. It may be that municipalities revealed as corrupt were more prone to experience different trends in homicide rates due to weak or inconsistent law enforcement, particularly with the presence of criminal groups in rural Mexico (Gutiérrez-Romero & Oviedo 2018, Depetris-Chauvin et al. 2022), or that corrupt municipalities, when aligned with the federal government, or revealed as corrupt in earlier time periods, experienced different trends due to partisan patronage (Sakurai & Theodoro 2020, Simpser et al. 2022, Depetris-Chauvin et al. 2022) and stronger priors (Chong et al. 2015, de Kadt & Lieberman 2020, Arias et al. 2022).

I divide the data by network strength quintiles for a clearer interpretation of heterogeneous effects. By dividing the data into quintiles, I can interpret the effect of corruption revelations on exam cheating at five different network strength levels, rather than a single average moderation effect coefficient. This partially mimics the use of a categorical variable interacted with the treatment variable, which would also grant greater interpretation capacity, but would be less intuitive given it uses as control units all localities regardless of network strength, thus comparing localities with strong networks to those with weak networks, despite their potentially different base characteristics. See Table 3 for an overview of the distribution of the average degree network variable after dividing it into quintiles.

Table 3: Network Strength Quintile Summary - Average Degree

Quintile	Observations	Minimum	Maximum	Mean	Median
1	29,769	1.000	2.095	1.772	1.804
2	29,769	2.095	2.597	2.345	2.345
3	29,768	2.597	3.102	2.843	2.839
4	29,768	3.102	3.765	3.417	3.410
5	29,768	3.765	7.763	4.534	4.344

Notes: These are the quintile cutoffs which determine the strength of the network variable, and the number of observations (grade-school-year) per quintile.

V. Results

A. Main Results

In Table 4, I show the results from my main specification in Equation 1. Here, I include the five models (quintiles), regressing exam cheating rates at the grade-school-year level, on my corruption (treatment) variable. In every model I include the full list of controls, including fixed-effects at the grade, school, municipality and year levels. I have also included the coefficient for 'Released', indicating the effect of having a report released on exam cheating rates, regardless of whether corruption in the FISM funds was found or not.

As shown in Table 4, the substantive effect of corruption on cheating increases notably in the higher network quintiles, particularly after the fourth quintile. This is consistent with my theory's expectations and with the statistical significance ($p\text{-value} < 0.1$) and sign of the main results in Ajzenman (2021). However, the statistical significance of these results is not consistent, so the sign and strength of these coefficients, particularly those in the first three quintiles, should be interpreted with caution.

Given the lack of statistical significance in the lower quintiles, I fail to reject the null that in

Table 4: Effect of Corruption on Cheating (Secondary Schools) by Network (Avg. Degree) Strength Distribution in Quintiles

	(Q1)	(Q2)	(Q3)	(Q4)	(Q5)
<i>Key Variables</i>					
ATT	-0.0051 (0.0079) [-9.6%]	-0.0095 (0.0159) [-17.9%]	-0.0041 (0.0123) [-7.7%]	0.0294* (0.0173) [55.5%]	0.0189* (0.0108) [35.7%]
Released	0.0040* (0.0023)	-0.0016 (0.0027)	0.0068** (0.0033)	0.0032 (0.0034)	0.0044 (0.0061)
<i>Controls</i>					
Political Controls	✓	✓	✓	✓	✓
Municipality Controls	✓	✓	✓	✓	✓
<i>Fixed Effects</i>					
Grade	✓	✓	✓	✓	✓
School	✓	✓	✓	✓	✓
Municipality	✓	✓	✓	✓	✓
Year	✓	✓	✓	✓	✓
<i>Model Fit</i>					
Observations	29,450	29,619	29,651	29,689	29,712
R ²	0.2513	0.2680	0.2653	0.2601	0.2558

Notes: Clustered standard errors in parentheses (municipality-level). Square brackets show the ATT effect as a percentage of the mean exam cheating rate (0.053). All regressions include grade, school, and year fixed effects. Political controls include incumbent party alignment, past corruption, and prior audits. Municipality controls include homicides, population, and fiscal indicators.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

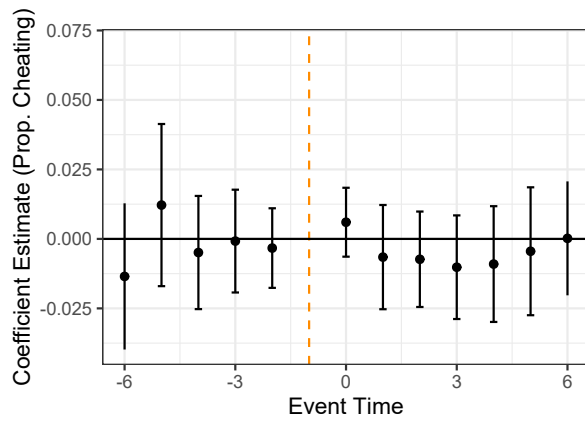
weaker locality-level family networks, when revealing corruption at the municipality level, secondary school students do not change their exam cheating rates (behaviour). The effects observed in the first three quintiles account for a decrease of 9.6% (Q1), 17.9% (Q2) and 7.7% (Q3) of the exam cheating rate mean in my sample. In my theory, I indicated that in rural localities, given their greater informational constraints relative to urban localities, it is more likely that social learning fails and that we observe null or even negative results. It may be that either the information about corruption did not reach students given the lack of effective informational pathways or that the weak networks did not permit coordination between students to cheat. In either case, these results suggest that when rural localities have weak family networks, the effects of corruption revelations on forms of behaviour like exam cheating are weaker than those in urban localities regardless of network strength, as reported in Ajzenman (2021).

The results from the fourth and fifth quintile regressions suggest that in stronger locality-level networks, when a municipality is reported to be corrupt in an audit, secondary school students increase their exam cheating rates by approximately 1.9-2.9 percentage points (pp). The effect observed represents a 35.7-55.5% increase in exam cheating rates from the exam cheating rate mean. This substantial increase in exam cheating rates suggests more pronounced behavioural changes than those reported by Ajzenman (2021) of 9.4-16.6%, thanks to the stronger family networks. The effect found here even surpasses that which he finds in high media exposure schools of 18.8-20.9%. This not only suggests further heterogeneity in his results but denotes the crucial role of networks in moderating the effect of corruption revelations on behaviour, similar to how information and the level of corruption were seen to moderate the effect of corruption revelations in Ajzenman's analysis.

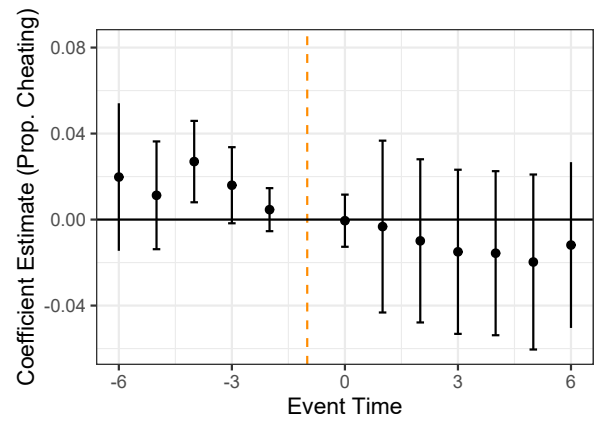
While the effects observed in Table 4 are encouraging given my theory's expectations, these results cannot be causally interpreted unless the identification assumptions are plausibly met. While we cannot directly test for the parallel trends or no anticipation assumptions, we can plot the *pre*-trends of our treated and controls units with respect to their exam cheating rates through event-study plots. These plots serve as visual falsification tests, which can make us slightly more or less confident of whether the parallel trends assumption holds in the case no or some statistically significant deviations are observed prior to treatment take-up, respectively.

B. Event-Study Plots

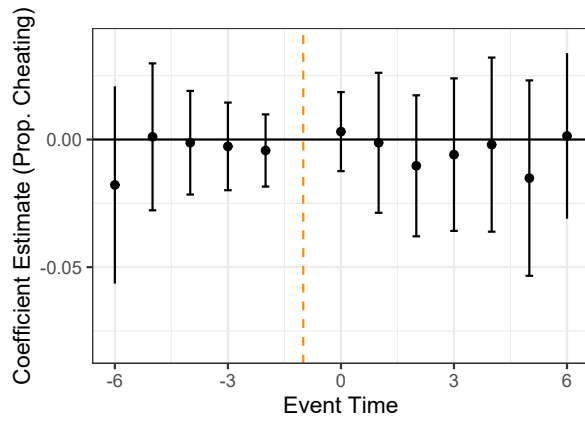
In Figure 3, we can see how there are no statistically significant deviations in the last period before treatment take-up for any of the quintiles. Nonetheless, there seem to be statistically significant deviations 3 and 4 years before treatment take-up in the fifth (Figure 3e) and second (Figure 3b) quintiles, respectively. While these pre-trend deviations are contained to just one year, these may signal that my results are not meeting the parallel trends assumption, or that some bias remains in the estimates. This is because, on average, units revealed as corrupt in the audits seemed to experience higher cheating rates preceding the release of the audit report. Given this, further tests are required to assess the validity of parallel trends, even if there are parallel pre-trends in the two years before the audits are released. Two tools stand out for this purpose: placebo (treatment timing) and equivalence tests.



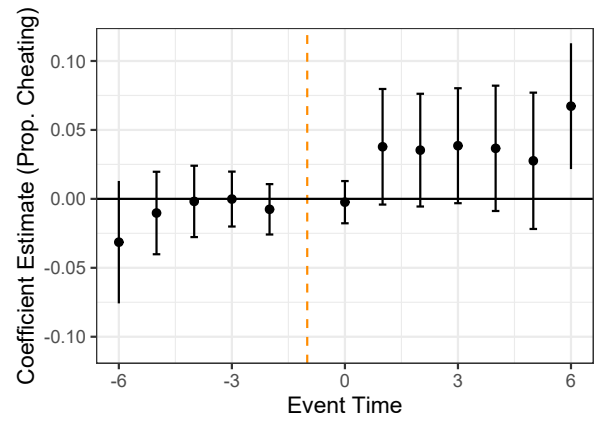
(a) Q1 Event Study



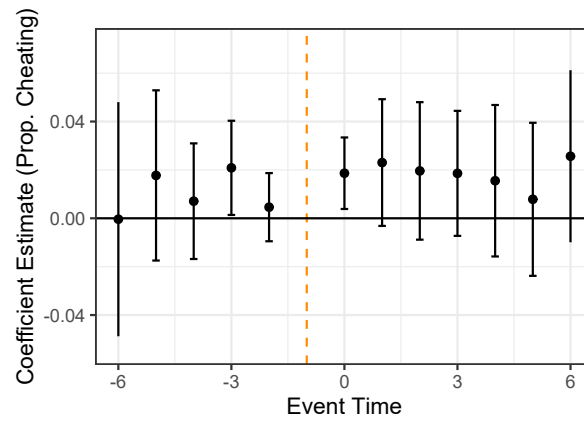
(b) Q2 Event Study



(c) Q3 Event Study



(d) Q4 Event Study



(e) Q5 Event Study

Figure 3: Event Study Results by Quintile Using Sun & Abraham (2021) Estimator

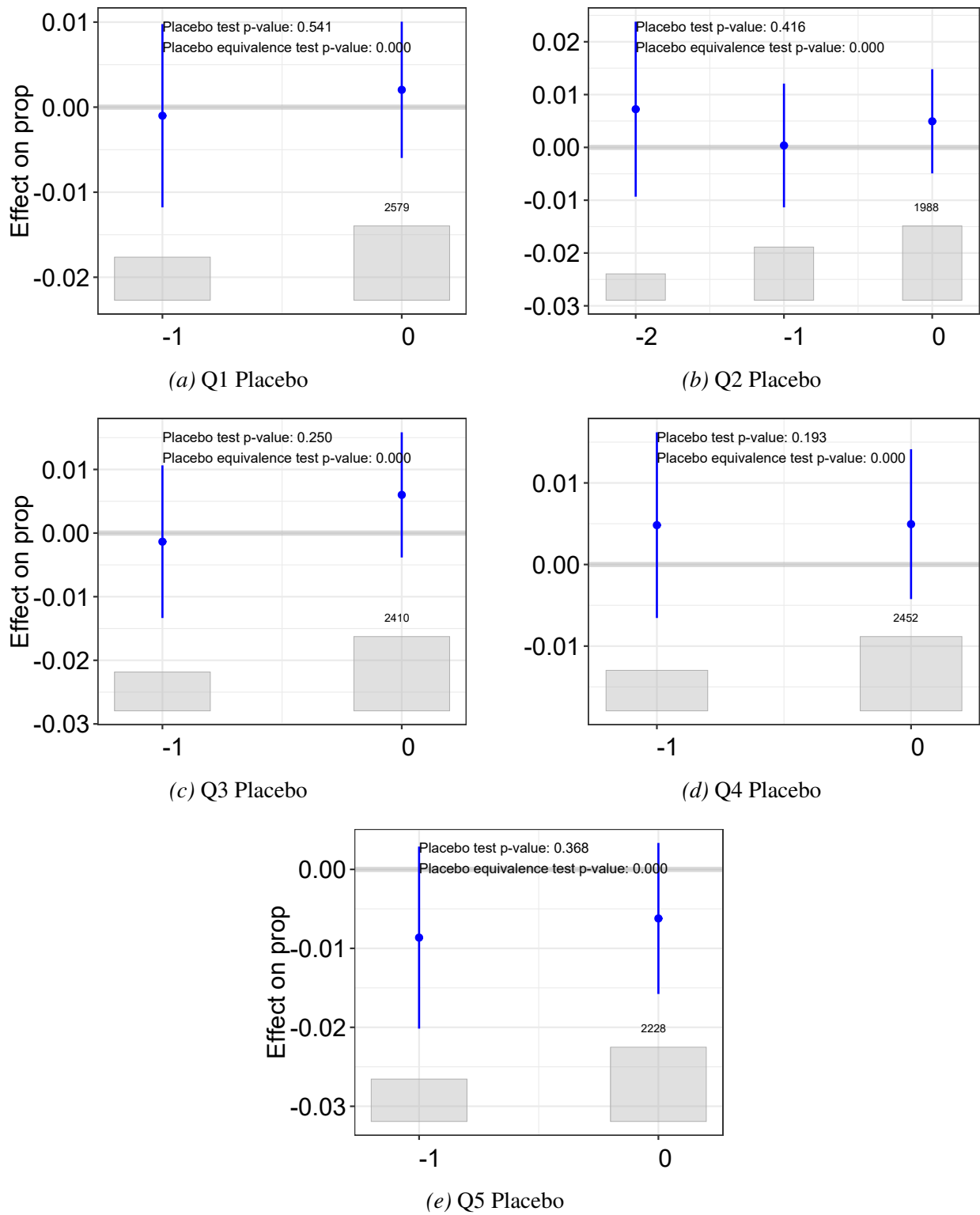


Figure 4: Placebo Treatment Timing (reference period = -3) Tests by Quintile using FEct estimator (Liu et al. 2024)

C. *Placebo and Equivalence Tests*

The placebo tests in Figure 4 present two values: a placebo test p-value, and a two-one-sided t-test (TOST) p-value, a type of equivalence test.⁷ Ideally, we want a high placebo test p-value (e.g. >5-10%), and a low TOST test p-value. We can see how in all my specifications (model by quintiles), neither the placebo test p-value nor TOST p-value indicate that parallel trends are violated. This grants the study greater confidence in believing the parallel trends assumption is met. Nonetheless, the (no) pre-trend test introduced by Liu et al. (2024) can help us further assess this key assumption.

In the equivalence tests presented in Figure 5, we are presented with two values: a goodness-of-fit test (a variant of the F-test) for zero residual averages in the pretreatment periods, and again the TOST test. In the F-test, we want higher values, as this suggests a better pre-trend fitting. All of my model specifications, across the three counterfactual estimators proposed by Liu et al. (2024), fail to indicate low F-test p-values, thus, from this, I fail to reject the null that there are *no* statistically significant deviations in the pretreatment periods. The interpretation of the TOST test p-value remains the same as in Figure 4 for the placebo tests, thus, from these falsification tests, I have no further reason to believe that the parallel trends assumption is not met for this specification. Overall, these tests while not conclusive, are informative, and suggest that the parallel trends assumption is met for this specification.

⁷See 'fect' R package documentation for a more comprehensive explanation of these tests.

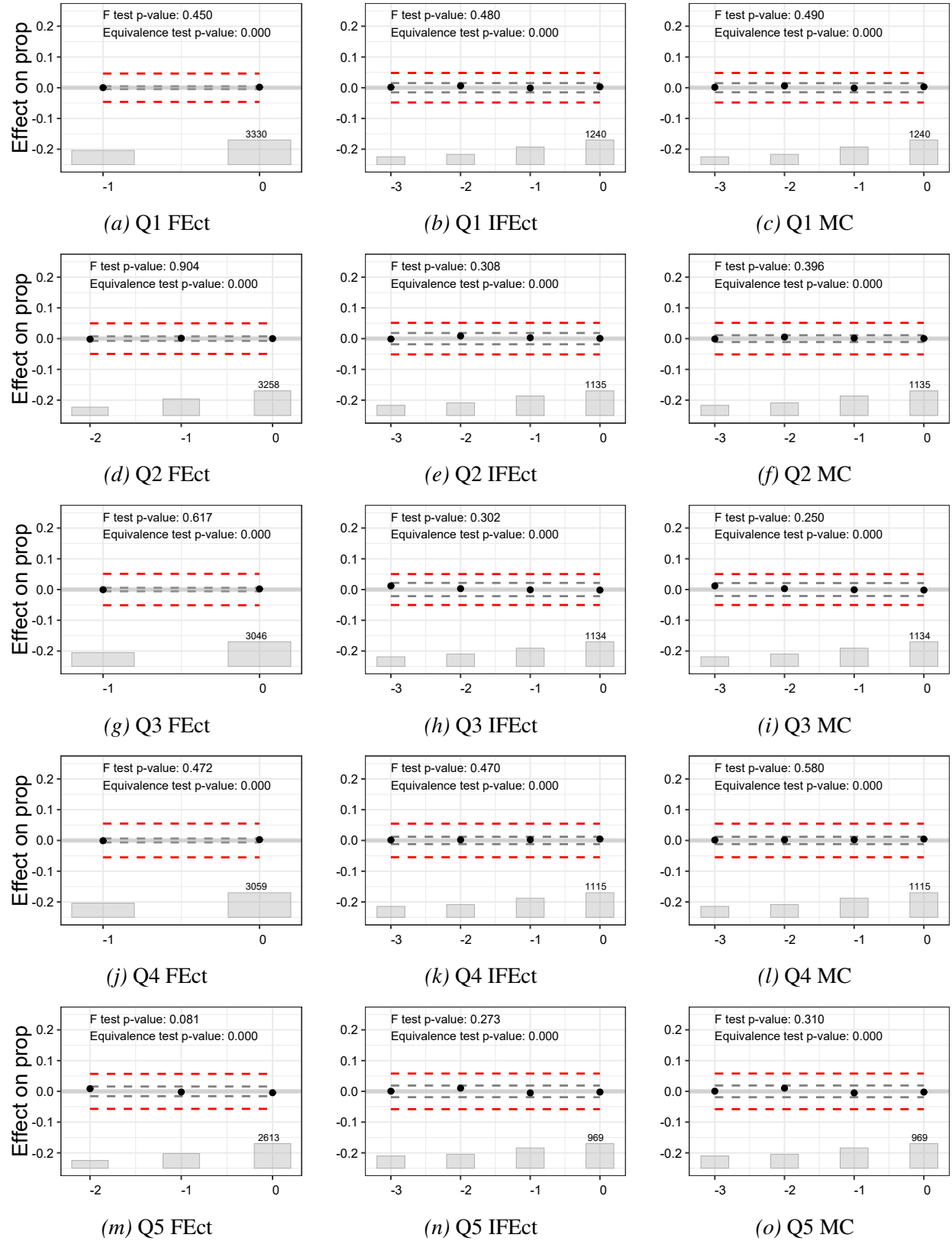


Figure 5: Equivalence Tests by Quintile using FECT, IFECT and MC estimators from Liu et al. (2024)

D. Robustness Tests

The results above point to a positive, strong, and statistically significant effect, at the 10% significance level, of corruption on exam cheating in rural localities but only when the locality-level family networks are in the upper two quintiles of the average degree distribution. While these results are encouraging and seem to support the theory presented above, a way of checking whether this effect is consistent is testing the effect of corruption on exam cheating but this time interacting it with the largest eigenvalue measure (see Appendix C for summary statistics and event-study plots). While there is a 93% correlation between these measures, they reflect slightly different things, as indicated in the Data section, so one cannot assume that the substantiveness or significance of the effects will mechanically hold.

As can be seen in Table 5, while the sign and strength of the effect of corruption on exam cheating hold, the statistical significance at the 10% level is lost in the upper two quintiles. This may be due to the conceptual differences between these two network measures, as average degree measures more closely reflect direct, local coordination rather than network-wide information "cascades". This mechanistic insight suggests that immediate peer effects matter more than being embedded in a globally central position. This may also suggest that strong family networks, rather than provide a network-wide cooperation channel through which information spreads effectively, lead to increased cheating when there is clear neighbour-to-neighbour (peer-to-peer) influence. This makes sense theoretically if one considers exam cheating as something that is not widely spoken about in the network but rather discussed in private. This also may suggest that for increased exam cheating, networks do not require coordination through central families, but just coordination between some families, regardless of their respective connectivity. Overall, this suggests that family net-

Table 5: Effect of Corruption on Cheating (Secondary Schools) - by Network (Largest Eigenvalue) Strength Distribution in Quintiles

	(Q1)	(Q2)	(Q3)	(Q4)	(Q5)
<i>Key Variables</i>					
ATT	-0.0067 (0.0076) [-12.6%]	-0.0097 (0.0187) [-18.3%]	0.0087 (0.0104) [16.4%]	0.0278 (0.0175) [52.5%]	0.0181 (0.0127) [34.2%]
Released	0.0037 (0.0024)	0.0020 (0.0029)	0.0020 (0.0030)	0.0078* (0.0044)	0.0006 (0.0054)
<i>Controls</i>					
Political Controls	✓	✓	✓	✓	✓
Municipality Controls	✓	✓	✓	✓	✓
<i>Fixed Effects</i>					
Grade	✓	✓	✓	✓	✓
School	✓	✓	✓	✓	✓
Municipality	✓	✓	✓	✓	✓
Year	✓	✓	✓	✓	✓
<i>Model Fit</i>					
Observations	29,426	29,609	29,667	29,699	29,720
R ²	0.24243	0.26991	0.26475	0.24667	0.27372

Notes: Clustered standard errors in parentheses (municipality-level). Square brackets show the ATT effect as a percentage of the mean exam cheating rate (0.053). All regressions include grade, school and year fixed effects. Political controls include incumbent party alignment, past corruption, and prior appointments. Municipality controls include crime, population, and fiscal indicators.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

works moderate the effect of corruption on exam cheating rates primarily through the coordination capacity mechanism rather than through the information transmission mechanism which is likely also present but to a lower degree in this context.

It is also important to note that the loss of statistical power is not as substantive or indicative of result fragility as it may seem at first glance, as the t-test value of the ATT coefficient in Q4 and Q5 results when changing from average degree to largest eigenvalue goes from 1.699 and 1.750 in Table 4 to 1.588 and 1.425 in Table 5, just over the 10% significance threshold. This highlights how the use of conventional significance thresholds to assert statistical significance overlooks the actual statistical power of results, often invalidating or validating results without a clear understanding of what changed between models.

Nonetheless, the variability in results across network measures motivates the use of an additional network measure to check whether significance or insignificance are more prevalent with this data and specification, such as Facebook's Social Connectedness Index (SCI). One of the problems with Facebook's SCI is that the only data this dissertation can rely on from it is that employed by Enríquez et al. (2024) for their experimental sample. This data is standardised, restricted to only rural localities to maintain interpretation consistency, and is at the municipality- rather than locality-level. The fact that the SCI data is at the municipality-level is not worrying given this network measure is fundamentally different from the family network data I built with the Prospera dataset, given it will not be as prone to the capture of spurious ties from more populous (less precise) administrative divisions.⁸ My dataset with SCI data is 23.66% of the sample used for the main results, or 35,212 observations. Given this, it seems reasonable to divide the network

⁸For a more complete explanation of Facebook's SCI, including methodology and validity, see Bailey et al. (2018).

connectedness data into two groups (i.e., high and low connectedness) rather than quintiles (see Appendix D for summary statistics and event-study plots). While this limits our interpretation into just above or below median values of the SCI, it does provide some valuable information, while maintaining statistical power. This slightly changes the empirical specification to:

$$Y_{g,s,t}^{(SCI)} = \kappa_{grade}^{(SCI)} + \alpha_{school}^{(SCI)} + \delta_{muni}^{(SCI)} + \lambda_t^{(SCI)} + \beta D_{muni,t}^{(SCI)} + \tau \mathbf{X}_{muni,t}^{(SCI)} + \epsilon_{school,t}^{(SCI)}, \quad SCI = High, Low \quad (2)$$

Notes: Equation 2 is estimated separately for above and below-median SCI values ($SCI \in \{High, Low\}$) of the distribution of Facebook’s Social Connectedness Index, aggregated at the municipality-level. All coefficients are allowed to vary from high- to low-SCI strength.

The results in Table 6 continue suggesting some network moderation effects, with municipalities with above-median SCI values experiencing a statistically significant (at the 10% level) and positive effect (3.35pp.) from corruption revelations on exam cheating, while municipalities with below-median SCI values do not experience a statistically significant effect from corruption revelations on exam cheating. These results strengthen the internal validity of our main results, and seem to support our theory on network moderation effects in rural localities. While in the discussion above, the lower prevalence of social media and internet use in rural localities was highlighted as an information transmission challenge relative to urban localities, the validity of Facebook’s SCI as a network measure comes from the evidence that shows that online ties closely reflect offline ties in contexts similar to the one studied here (Bisbee & Larson 2017, Enríquez et al. 2024). From the event-study plots (see Appendix D), it seems plausible that the parallel trends assumption is

Table 6: Effect of Corruption on Cheating (Secondary Schools) - by Network (Facebook SCI) Distribution

	Low-SCI	High-SCI
<i>Key Variables</i>		
ATT	0.0112 (0.0171) [21.1%]	0.0335* (0.0185) [63.2%]
Released	0.0025 (0.0083)	0.0037 (0.0064)
<i>Controls</i>		
Political Controls	✓	✓
Municipality Controls	✓	✓
<i>Fixed Effects</i>		
Grade	✓	✓
School	✓	✓
Municipality	✓	✓
Year	✓	✓
<i>Model Fit</i>		
Observations	17,511	17,540
R ²	0.26360	0.27659

Notes: Clustered standard errors in parentheses (municipality-level). Square brackets show the ATT effect as a percentage of the mean exam cheating rate (0.053). All regressions include grade, school and year fixed effects. Political controls include incumbent party alignment, past corruption, and prior appointments. Municipality controls include crime, population, and fiscal indicators.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

also met for this empirical specification, as there are no statistically significant deviations in the pre-trends.

An additional robustness test of interest to this study is checking whether the strength and significance of the main results depend on the cut-off points used. In Table 7, I present the main results when splitting my network (average degree) variable into quartiles, and in Table 8, I present the main results when splitting the network variable into tertiles. See Appendix E for summary statistics and event-study plots.

Table 7: Effect of Corruption on Cheating (Secondary Schools) by Network (Avg. Degree) Quartile Distribution

	(Q1)	(Q2)	(Q3)	(Q4)
<i>Key Variables</i>				
ATT	-0.0087 (0.0087) [-16.4%]	-0.0036 (0.0094) [-6.8%]	0.0263* (0.0154) [49.6%]	0.0164 (0.0111) [31.0%]
Released	0.0024 (0.0019)	0.0008 (0.0026)	0.0058** (0.0029)	0.0045 (0.0058)
<i>Controls</i>				
Political Controls	✓	✓	✓	✓
Municipality Controls	✓	✓	✓	✓
<i>Fixed Effects</i>				
Grade	✓	✓	✓	✓
School	✓	✓	✓	✓
Municipality	✓	✓	✓	✓
Year	✓	✓	✓	✓
<i>Model Fit</i>				
Observations	36,847	37,043	37,095	37,137
R ²	0.2499	0.2730	0.2584	0.2564

Notes: Clustered standard errors in parentheses (municipality-level). Square brackets show the ATT effect as a percentage of the mean exam cheating rate (0.053). All regressions include grade, school, and year fixed effects. Political controls include incumbent party alignment, past corruption, and prior appointments. Municipality controls include crime, population, and fiscal indicators.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table 8: Effect of Corruption on Cheating (Secondary Schools) by Network (Avg. Degree) Tertile Distribution

	(T1)	(T2)	(T3)
<i>Key Variables</i>			
ATT	-0.0071 (0.0088) [-13.4%]	-0.0009 (0.0104) [-1.7%]	0.0278** (0.0134) [52.5%]
Released	0.0003 (0.0020)	0.0068** (0.0027)	0.0031 (0.0050)
<i>Controls</i>			
Political Controls	✓	✓	✓
Municipality Controls	✓	✓	✓
<i>Fixed Effects</i>			
Grade	✓	✓	✓
School	✓	✓	✓
Municipality	✓	✓	✓
Year	✓	✓	✓
<i>Model Fit</i>			
Observations	49,179	49,432	49,512
R ²	0.2577	0.2625	0.2580

Notes: Clustered standard errors in parentheses (municipality-level). Square brackets show the ATT effect as a percentage of the mean exam cheating rate (0.053). All regressions include grade, school, and year fixed effects. Political controls include incumbent party alignment, past corruption, and prior appointments. Municipality controls include crime, population, and fiscal indicators.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

In both Tables, the sign and strength of results remain close to those presented in Table 4, however, there is a loss of statistical significance in the fourth quartile of Table 7, suggesting that network strength moderation effects are non-monotonic, with a subtle drop in strength and significance in the stronger networks. While this drop in strength was already visible in the fifth quintile of the main analysis, the loss of significance at the 10% level in the fourth quartile should warrant certain caution into assuming that stronger networks will inevitably lead to greater information transmission and coordination. It may be that alternative mechanisms are driving down the effect in very strong networks, such as stronger cheating or corruption priors, or even a level of collusion in the corruption scandals. In Appendix E one can see how the parallel trends assumption also seems to be met from the lack of statistically significant pre-treatment deviations.

VI. Limitations

First off, the results observed only hold for the sample used, which is substantively different from the sample used in Ajzenman (2021). It may be that family network moderation effects do not hold in urban localities as student-to-student cheating coordination may be driven primarily through friends and not family connections. It may also be, that the effects observed are not representative of the entirety of rural Mexico either, as I could not incorporate network data for rural localities without schools, only those which had a school in their locality in Ajzenman's sample. It may be that the actual moderation effects are weaker, or stronger, however, it is unreasonable to claim anything else without incorporating geospatial data which considers the family network strength of neighbouring localities on top of the network strength of the school's locality. The role of networks on behavioural change may remain fairly similar in urban communities, even if characterised in a different way. A suggested area for further research would be to exploit Facebook's SCI micro-level data in the US to explore a similar research question in urban communities as well, given the SCI will more likely capture real, urban social ties.

One further limitation of this study comes from potential bias in my network construction method. While various steps were taken to minimise non-classical network measurement error, such as using rural-only localities, aggregating at locality-level, and using an administrative dataset rather than sampled network data, it may be that the usage of the Prospera dataset for network construction biased my measures. It may be that my network measures only closely represent the family network strength of poor individuals within a locality, not the family network strength of the entire locality. Still, the previous use of the Prospera dataset for network construction in papers with sim-

ilar contexts (Arias et al. 2019) and the network-construction benefits of administrative datasets over sampled network data (Chandrasekhar & Lewis 2016, Chandrasekhar et al. 2018, Alt et al. 2022) make me confident that the measures used here are valid. The effects observed with Facebook's SCI were consistent with those captured in my main analysis, and highlight how even when using different network measures, the results hold. A suggested improvement for future research would be to construct the network data using individuals as nodes for robustness, and exploiting other administrative datasets to construct the network measures so the likelihood of bias from an unrepresentative network "sample" is reduced.

Another limitation in my analysis comes from not being able to confirm whether the parallel trends assumption is met or not. While the event-study plots, placebo tests, and equivalence tests seemed to suggest that the assumption was met, there is no method which can give us 'proof' of whether it is met or not (Baker et al. 2025). In the case there were parallel trends violations, it is likely that my results would not remain the same. Conducting a sensitivity analysis like that proposed by Rambachan & Roth (2023), which induces a level of bias into the results to check how large of a deviation from the parallel trends assumption would change results, will almost surely lead to non-significant results, as reported by Chiu et al. (2025). It is improbable that results remain significant when there is a violation of parallel trends even when their significance is at the 5% or 1% level, so mine would probably turn insignificant given I only have statistical significance at the 10% significance level. Nonetheless, as explained above, the tests conducted, while not conclusive, were informative and consistent, thus lead me to believe that the parallel trends assumption is met for this specification.

Another methodological limitation from this study, is that I cannot be certain that the Stable Unit

Treatment Value Assumption (or SUTVA) is met. This limitation is particularly relevant given my treatment variable is binary, and thus hides the actual extent of corruption revealed (i.e., percentage of funds unaccounted for in audits), so treatment is not necessarily consistent across units, especially when one considers the role of priors (Chong et al. 2015, Arias et al. 2022). Unfortunately, no method currently exists which can account for staggered (more than two periods) and continuous treatments, as well as absorbing and non-absorbing ones, as the treatment used here could be studied in both forms. Ajzenman (2021) reported similar results when only considering as treated localities with a higher level of corruption in the audits, so it is fair to assume that this would also be evidenced here, even if there was no more space to check for it. It is important to note, however, that the literature on difference-in-differences estimators is concurrently expanding (de Chaisemartin & D'Haultfoeuille 2020, Imai et al. 2023, Callaway et al. 2024, de Chaisemartin et al. 2024, Liu et al. 2024), so new methods which can account for this may appear soon, enabling a reappraisal of the results.

One of the bigger limitations in my analysis comes from the fact that we cannot confidently separate the prevalence of either network moderation mechanism: information transmission and coordination capacity. To identify the prevalence of either mechanism, I would need greater micro-level data which can reliably separate the informational and coordination effects of strong family networks. The loss of statistical power in the analysis using largest eigenvalue measures may suggest that strong family networks drive higher cheating rates through direct communication rather than through greater network centrality, potentially indicating that coordination is more prevalent than information transmission, but again it is non-sensible to confidently claim this with just my current results.

A suggested improvement for future research would be to exploit media access variation by locality, as is done in Larreguy et al. (2020), to explore whether network moderation effects are sustained when controlling or interacting for media access (e.g., radio ownership, radio supply). It may be that when there is already high media exposure, the network moderation effects disappear, and this would indicate that networks mostly work as an information transmission mechanism. Alternatively, if network moderation effects are well-sustained when there is high media access, this suggests that networks impact cheating rates following corruption revelations primarily through the coordination capacity mechanism.

VII. Conclusion

This study has shown there are large, statistically significant effects of corruption revelations on exam cheating rates in rural localities but only in the stronger network quintiles, with some variability in magnitude and significance for the strongest locality-level networks. As indicated in Dunning et al. (2019), null effects in the lower quintiles are informative of the challenges that rural localities in developing contexts experience when presented with information on malfeasant incumbents.

The differences in effect magnitude in the higher quintiles relative to Ajzenman (2021) suggest that strong family networks can not only mitigate some of the lost informational effectiveness of non-competitive media markets and other conditions in rural communities, but that they can revert the lost effect power relative to urban localities. The effects observed are substantively large and have important consequences for local development given the implications school cheating has for broader rule compliance and local educational achievement.

This study has helped contribute to the literature on family networks as both information transmission and coordination mechanisms, showing how in more strongly connected networks, school children react more strongly to political information. While this finding only applies to our case of rural Mexico, one could imagine similar effects in other contexts where politics is primarily discussed within the household. It seems probable that other characterisations of social networks would continue moderating these effects, such as friendship ties.

This dissertation has important policy implications, ranging from the importance of effective information transmission in rural localities, to the consequences of corruption on educational outcomes.

This study has helped extend the literature on the behavioural spillovers of corruption, showing how even in information-constrained environments there can be socially-detrimental consequences from corruption such as increased exam cheating, a form of decreased rule compliance. Accountability should be strengthened to ensure that corrupt officials do not further erode compliance and educational quality. While there may be certain contexts in which increasing formal accountability may limit the informativeness of government outcomes like those presented in audits (Ashworth et al. 2017), the results from this study suggest that alternative information transmission mechanisms like family networks can facilitate behavioural, coordinated responses, as shown in Arias et al. (2019). The finding that family networks can mitigate some of the informational constraints faced in rural communities should be taken into consideration by policymakers and researchers who can try leveraging these networks to push for and study effective anti-corruption reforms.

This study has reiterated the need for more effective information transmission channels in rural areas with less competitive media markets and lower internet availability. The lack of competitive media markets at least partially drives the lack of a statistically significant effect in the lower quintiles of the network distribution, and likely contributes to other developmental and accountability issues in rural areas of Mexico.

Future studies on this topic should attempt to solve some of the limitations suggested above, while also trying to extend the literature on behavioural spillovers of corruption to other areas of rule compliance such as tax evasion, public health mandates, or disorderly conduct. If one believes that early life experiences and attitudes can shape our later life behaviour, it is plausible that the normalisation and increased prevalence of cheating and dishonesty following corruption could lead to other, more detrimental forms of rule defiance later in life. Corruption is detrimental to

development, and this study has shown just one of the pathways through which it undermines societal progress.

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Appendix

A Replication Materials

All data preparation scripts, analysis code, and documentation necessary to replicate the results in this dissertation are available in an anonymised GitHub repository at:

<https://github.com/candidate-40284/Replication-Package-MSc-Dissertation-2024-25>

The repository includes:

1. R scripts for all main analyses and robustness checks
2. Network construction code
3. Documentation of data sources and variable definitions
4. Instructions for replicating all tables and figures

B Network Analysis Summary Plots

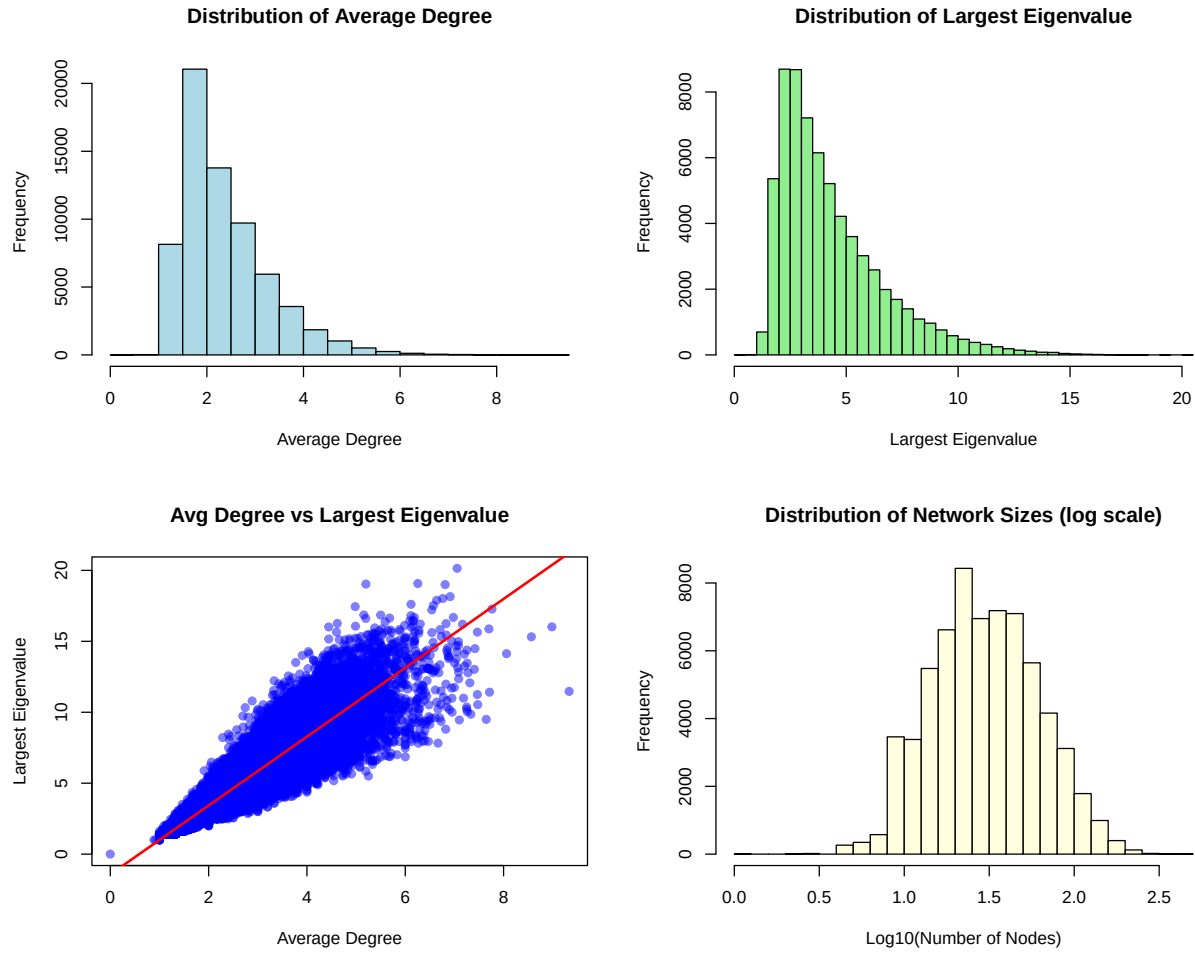


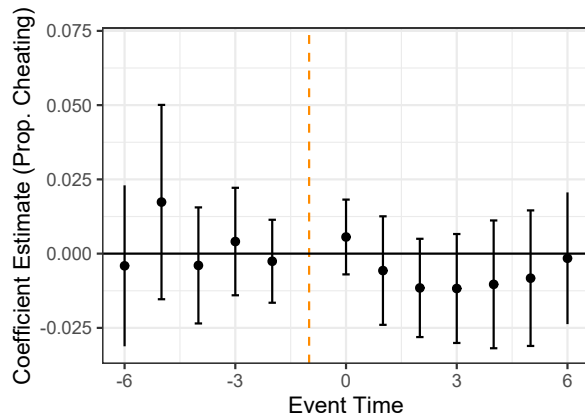
Figure B.1: Distribution of Average Degree and Largest Eigenvalue network measures at the locality-level and using families as nodes. Correlation graph of Average Degree (x-axis) and Largest Eigenvalue (y-axis) network measures. Distribution of network sizes used in dataset.

C Largest Eigenvalue Summary by Quintile and Event-Study Plots

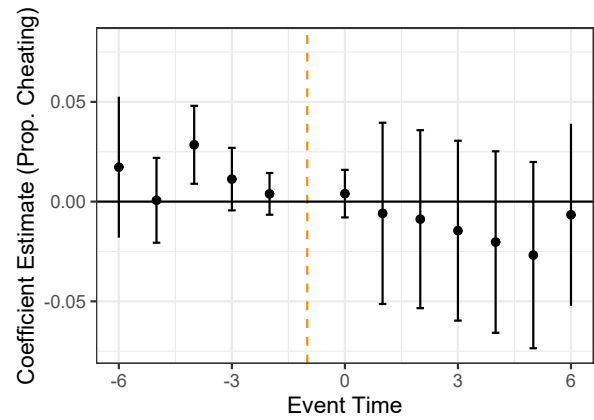
Table C.1: Network Strength Quintile Summary - Largest Eigenvalue

Quintile	Observations	Minimum	Maximum	Mean	Median
1	29,769	1.000	3.738	2.921	3.001
2	29,769	3.738	5.234	4.474	4.461
3	29,768	5.234	6.705	5.953	5.937
4	29,768	6.705	8.661	7.626	7.608
5	29,768	8.661	19.073	10.664	10.160

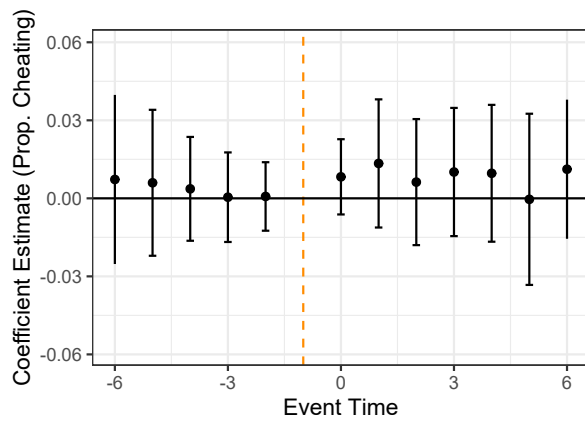
Notes: These are the quintile cutoffs which determine the strength of the network variable, and the number of observations (grade-school-year) per quintile.



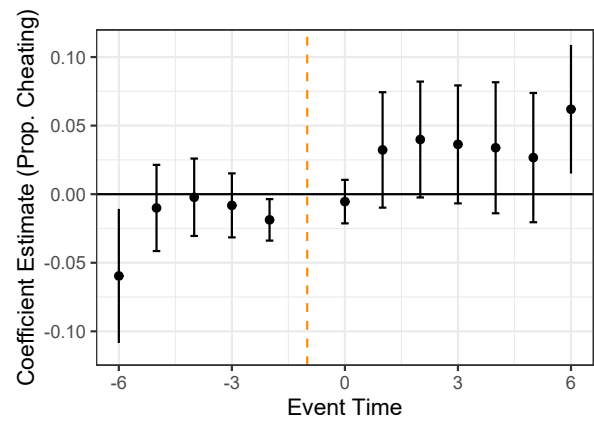
(a) Q1 Event Study



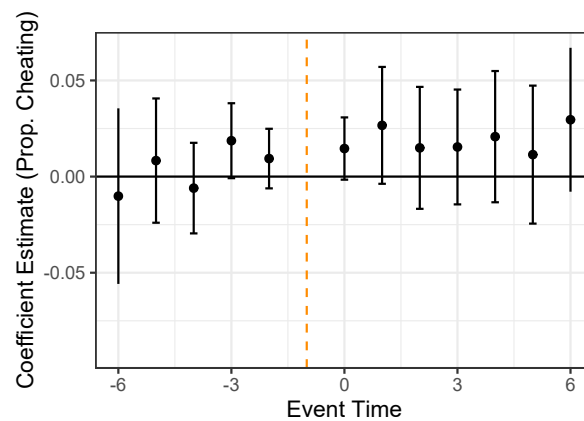
(b) Q2 Event Study



(c) Q3 Event Study



(d) Q4 Event Study



(e) Q5 Event Study

Figure C.1: Event Study Results by Quintile (Largest Eigenvalue) Using Sun & Abraham (2021) Estimator

D Facebook's Social Connectedness Index (SCI) Summary and Event-Study Plots

Table D.1: Network Strength Summary - Facebook's SCI

Strength	Observations	Minimum	Maximum	Mean	Median
Low	17,606	-0.595	-0.368	-0.503	-0.508
High	17,606	-0.368	5.008	0.201	-0.015

Notes: These are the cutoffs which determine the strength of the SCI network variable, and the number of observations (grade-school-year) per group.

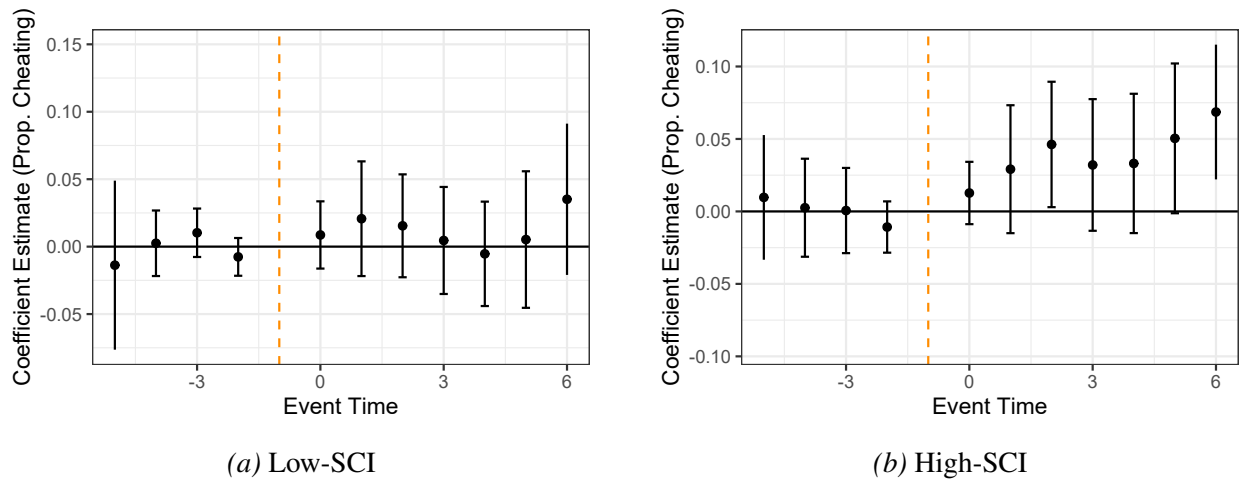


Figure D.1: Event Study Results by SCI Level

E Average Degree Summary by Quartile and Tertile and Event-Study Plots

Table E.1: Network Strength Quartile Summary - Average Degree

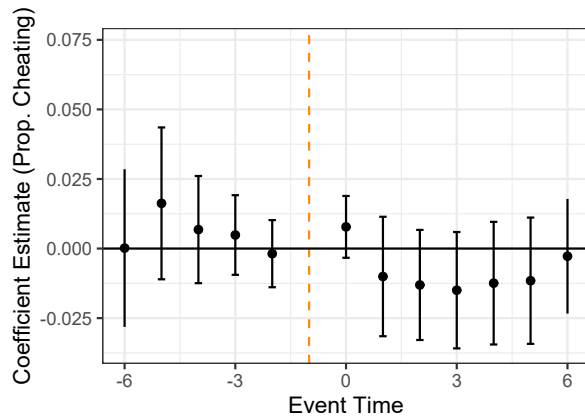
Quartile	Observations	Minimum	Maximum	Mean	Median
1	37,211	1.000	2.216	1.849	1.887
2	37,211	2.216	2.839	2.532	2.532
3	37,211	2.839	3.574	3.187	3.175
4	37,211	3.574	7.763	4.360	4.163

Notes: These are the quartile cutoffs which determine the strength of the network variable, and the number of observations (grade-school-year) per quartile.

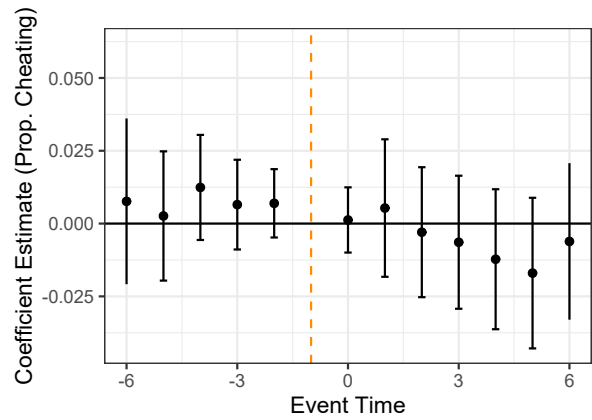
Table E.2: Network Strength Tertile Summary - Average Degree

Tertile	Observations	Minimum	Maximum	Mean	Median
1	49,614	1.000	2.429	1.968	2.000
2	49,614	2.429	3.308	2.848	2.839
3	49,614	3.308	7.763	4.130	3.926

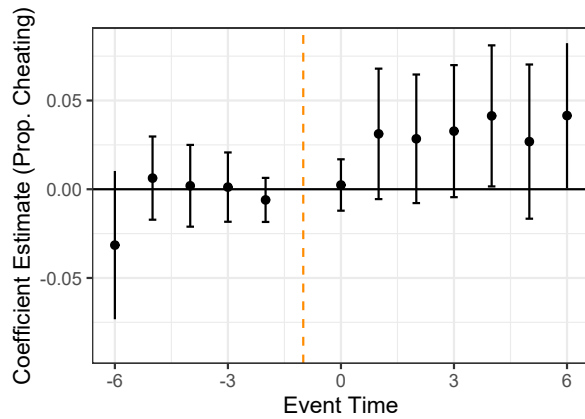
Notes: These are the tertile cutoffs which determine the strength of the network variable, and the number of observations (grade-school-year) per tertile.



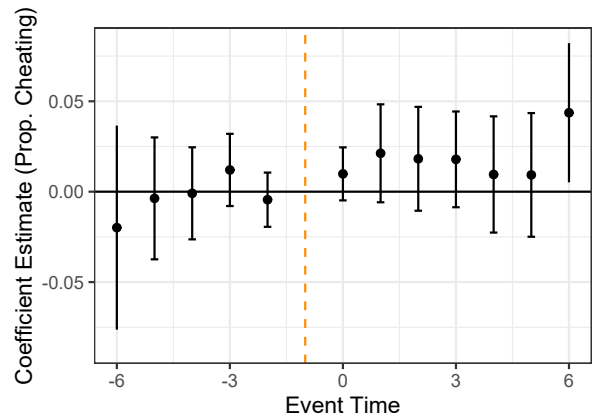
(a) Q1 Event Study



(b) Q2 Event Study

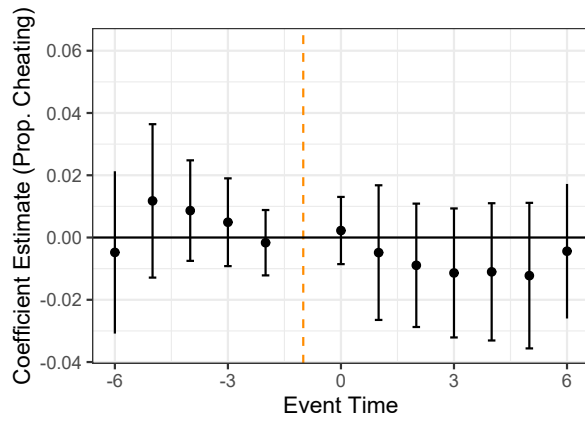


(c) Q3 Event Study

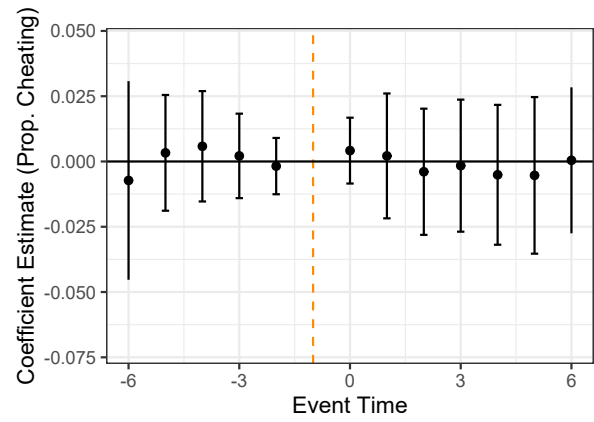


(d) Q4 Event Study

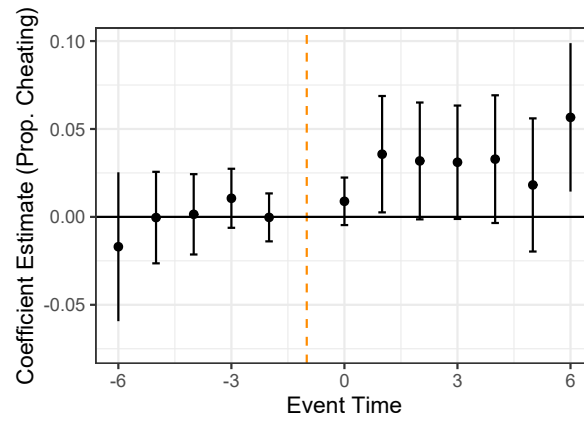
Figure E.1: Event Study Results by Quartile Using Sun & Abraham (2021) Estimator



(a) T1 Event Study



(b) T2 Event Study



(c) T3 Event Study

Figure E.2: Event Study Results by Tertile Using Sun & Abraham (2021) Estimator